

Part 6

Q8:

~~$-3 + 0.6x = 0$~~

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$$0.6x = 3 \quad | : 0.6$$

$p = x = 5$

$$Z = -3 + 0.6 \cdot 3 = -1.2$$

~~$1 - 0.231 = 0.769$~~

$$p = \frac{1}{1 + 3.32} = 0.231$$

∴

$$\frac{1}{1 + 3.32} = 0.231$$

$$0.231 \leq 0.5$$

Q9

$$Z = -3 + 0.6 \cdot 8 = 1.8$$

$$P = \frac{1}{1 + e^{-(1.8)}}$$

$\Downarrow$

$$\frac{1}{1.165}$$

$$= 0.858$$

80% profit

100% sure

$$X = 7.31 \text{ (req)}$$

(1)

(2)

Bonus

$$0.8 = \frac{1}{1 + e^{-(x)}}$$

$$\frac{1}{0.8} = 1 + e^{-(x)}$$

$$1.25 = 1 + e^{-(x)}$$

$$0.25 = e^{-(x)}$$

$$\ln(0.25) = -x$$

$$1.386 = x$$

$$-3 + 0.6x = 1.386$$

$$x = 7.31$$